Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/08/2020
PI Name:	Pecaut, Michael Ph.D.		
Project Title:	Impact of Spaceflight on Primary and Secondary Antibody Responses		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	mpecaut@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8372
Organization Name:	Loma Linda University		
PI Address 1:	Division of Biomedical Engineering Sciences, Department of Basic Sciences		
PI Address 2:	11175 Campus St, Chan Shun Pavilion, A-1016		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	07/31/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/21) NOTE: End date changed to 3/30/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 09/30/2020 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 11/30/2018 per NSSC (Ed., 10/10/17) NOTE: End date changed to 11/30/2017 per NSSC (Ed., 6/3/16)		
Key Personnel Changes/Previous PI:	June 2019 report: Co-Investigator Stephen K. Chapes, Ph.D., retired from his position at Kansas State University (KSU) and is now serving as a CoI consultant on this grant. September 2018 report: Fomer CoI Daila Gridley, Ph.D., is no longer participating in the project.		
COI Name (Institution):	Chapes, Stephen Ph.D. (Retired (formerly from Kansas State University))		
Grant/Contract No.:	NNX13AN34G		
Performance Goal No.:			
Performance Goal Text:			

	The spaceflight environment influences immune cell distribution, ex vivo responses to antigen, and cytokine expression. However, few studies have characterized the impact of the spaceflight environment on the host immune response following an in vivo immune challenge, particularly those involving B cells and antibody production. The International Space Station (ISS) provides an ideal laboratory within which investigators can explore this unique environment and its impact on the host response. Through the use of a mouse model (with an immune system that closely parallels that of humans), we will establish a link between exposure to the spaceflight environment and decrements in immune system activation. This study is significant because it will directly test the hypothesis that host immunological health will be compromised in space. On-orbit Flight Activities, Procedures, and Timeline Mice will receive health checks upon unloading and transfer from the Transporter to the Habitat after capsule docking no later than 24 hrs after hatch opening. Animal health will then be monitored daily via continuous video recording from one camera at a time from all of the Habitats on-orbit (including other investigations which may be running concurrently) and down-linked to Ames science staff. Ground personnel shall have the option of programming or manually controlling which camera is being down-linked and recorded at any given time. Water levels will be checked every 3-4 days. After a minimum of 14 days after animal transfer from the Transporter to the Habitat, all 20 mice will be moved to the MSG and receive intraperitoneal (i.p.) injection of vehicle (5 mice), Tetanus Toxoid (TT) vaccine (5 mice), CpG ODN adjuvant (5 mice), or TT vaccine + CpG ODN adjuvant (5 mice). The dose of TT and/or CpG ODN will be 10 microg/ mouse and 40 microg/ mouse, respectively. Pre-mixed doses will be prepared preflight in 100 microl saline and stored at -20 to -30°C until the appropriate time point. All injections and blood draws will use a 1 mL 26G 3/8” tuberculin syringe. All injections will be i.p. Before use, the CpG adjuvant must be stored at -20°C or colder until use and the TT vaccine must be stored at 4°C. At 14 +/- 2 days after injection, mice will be transferred to the MSG and sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Following spleen collection, the remains of the mouse carcass will be wrapped in tin foil, put in a zip-lock bags and placed in a -80°C environment (or colder) to preserve the tissues for future biospecimen sharing. Blood must not be allowed to sit at ambient temperature for longer than 1 hour post-collection; if operational requirements require the blood collected to be stored in excess of 1 hour before centrifugation, the blood must be transferred to a 4°C environment until centrifugation, for a period not to exceed 4 hours. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to Earth. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable. <table border=1 cellspacing=0 cellpadding=0 align=center width=672> <tr > <td width=91> Time </td> <td width=252> Activity & Estimated Time Required </td> <td width=76 > Thermal Req. </td> <td width=253 > Comments, Location & Special Considerations </td> </tr> <tr > <td width=91 > Launch to L + 45 days </td> <td width=252 > Animal health checks via video monitoring; water level checks every 3-4 days </td> <td width=76 > 20 – 30°C </td> <td width=253 > Video downlink required for health checks </td> </tr> <tr> <td width=91 > Transfer to Habitat + 14 d </td> <td width=252 > All 20 mice will be transferred to habitat </td> <td width=76 > 20 – 30°C </td> <td width=253 > Requires transfer to and from Habitat with MTB; </td> </tr> <tr > <td width=91 > Injection + 14d +/- 2 d </td> <td width=252 > At specific time points, all 20 mice will be removed from AEM and given appropriate injection </td> <td width=76 > 20 – 30°C / -80°C </td> <td width=253 > Required transfer to MSG for injections </td> </tr> <tr > <td width=91 > Dissection + 0 d </td> <td width=252 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen. Blood and spleen samples kept at -80^oC until delivered to laboratory of PI </td> <td width=76 > -80°C </td> <td width=253 > Required transfer to MSG for dissection, processing of blood in the centrifuge, and storage of samples at -80°C for up to 6 months. </td> </tr> </table> Hardware Housekeeping Operations: Crew-performed Daily hardware checks will occur. Habitats will be kept at a regular 12hr:12hr day:night light cycle to match the day/night programmed to be synchronous with the crew’s day/night cycle upon arrival. Water refill of habitats shall be performed. To prevent rodent food bars from becoming too hard or soiled, they will be replaced with a fresh food bar plate (from 4°C storage) at a time to not exceed 30 days after the introduction of mice to a Habitat.
Task Description:	Ground Activities Accompanying On-orbit Activities and Timeline Ground controls will be run following spaceflight environmental parameters in orbital environmental simulator(s) (OES) chambers at KSC ACF. Delay between launch and start of environmental controls may be up to 8 days, depending on the logistics and capability of the telemetry system. Mice will be housed in flight Habitats with RFB diet and lixit water to match on-orbit groups. The operations done on ground will be conducted to match the same relative time points during the day/night cycle of the animals on-orbit, so that procedures on ground control mice are conducted at the same points in the animals’ circadian rhythm. Each experimental group for ground controls will be doubled to improve the statistical significance of the experiment with a total of 40 mice (4 groups of 10). Therefore, 2 Transporters will be required for the flight in orbital environmental simulator(s) (OES) that have been programmed with telemetry data from the flight. At the same relative time point (accounting for delay in starting ground controls after flight) the mice are transferred from the Transporter to the Habitat, the 40 ground control mice will be transferred from the 2 Transporters into 4 Habitats, and maintained in OES programmed with telemetry data from the ISS. OES shall be programmed to mimic the following conditions experienced by the Transport and then the Habitat: CO2, O2, temperature, and humidity. Injection and dissection must be performed as close as possible to how they are done on-orbit. In particular, the time of day that mice are sacrificed and dissected on the ground should match the same point in the circadian rhythms of the mice on-orbit. However, dissections or ground control mice can be conducted on standard laboratory benches/tables

with chairs that are more ergonomic to the operators. For ground controls, dissection pins or needles may be used to pin down mice for dissection, as can spray bottles of wetting the mice before dissection. Flight hardware is required when it may affect the tissue dissection or preservation, such as having flight cold blocks in mini cold bags for tissue freezing.

Mice will receive health checks daily via video in the Habitat and water levels will be checked every 3-4 days. After a minimum of 14 days in the OES, all 40 mice will receive an i.p. injection of vehicle (10 mice), TT vaccine (10 mice), CpG adjuvant (10 mice), or TT vaccine + CpG adjuvant (10 mice). At 14 +/- 2 days after injection, mice will be sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Duration for which blood is stored at ambient or 4°C shall be matched as closely as possible to the timelines the same procedures will be performed on-orbit. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to the laboratory of the PI. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.

<table class=MsoNormalTable border=1 cellspacing=0 cellpadding=0 align=center width=672 > <tr > <td width=145 > Time </td> <td width=186 > Activity </td> <td width=66 > Thermal Req. </td> <td width=275 > Comments, Justification, Special Considerations </td> </tr> <tr > <td width=145 > Beginning of ground controls to +45 days </td> <td width=186 > Animal health checks; water level checks every 3-4 days </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Video downlink required for health checks </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > At specific time points, all 40 mice will be removed from AEM and given appropriate injection </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Transfer from cages to a regular lab bench and back w/ standard cages. </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen </td> <td width=66 > -80^oC </td> <td width=275 > Processing of blood in a centrifuge, and storage of samples at -80^oC </td> </tr> </table>

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

According to the National Institute of Aging (NIA), “Life expectancy nearly doubled during the 20th century with a ten-fold increase in the number of Americans age 65 or older.” Furthermore, “there are approximately 35 million Americans age 65 or older, and this number is expected to double in the next 25 years.” Consequently, there is a rapidly growing need to understand the mechanisms underlying age-associated illness. Indeed, in the last decade, research has repeatedly demonstrated an impact of aging on immune function. Low-grade increases in circulating inflammatory cytokine secretion are often associated with senescence. Furthermore, studies suggest that these disruptions are likely due to decrements in cellular repair mechanisms and/or increases in antigenic load. Given this disruption in immune function, it is perhaps not surprising that many of the diseases associated with aging — such as type 2 diabetes, rheumatoid arthritis, Parkinson’s disease, Alzheimer’s disease, cardiovascular disease — have strong inflammatory underpinnings.
There are strong indications that the physiological effects of spaceflight are also similar to those seen in senescence. For instance, evidence suggests that age-associated enhancements in inflammatory cytokines are linked to changes in adipose tissue and energy balance. Similar links between metabolism, adipose tissue, and inflammation have also been shown to manifest in astronauts and animal models after exposure to spaceflight. We propose to use long-term exposure to the spaceflight environment to model the effects of aging on immune function. Although these initial studies involve mice, the next obvious step in this work will be to work directly with astronauts. Because astronauts tend to be extremely healthy, these effects will not be influenced by confounding factors such as clinically relevant, age-related disease. Also, because the effects of spaceflight on immunity are temporary, generally returning to baseline levels post-flight, this will ultimately give us some insight into whether or not the effects of aging are, indeed, reversible.

Although many different aspects of immune function are influenced by both aging and spaceflight, these experiments will focus on B cells and the generation of antibodies. Specifically, we will use next generation RNA sequencing techniques to characterize the B cell Ig heavy chain in animals that are challenged with tetanus toxoid. The Ig heavy chain is the component of antibodies responsible for identifying specific sections of the tetanus toxoid, thereby coordinating subsequent immune responses. Comparisons will be made between the B cell repertoires generated on ISS and on Earth. The changes in B cell Ig heavy chain usage will provide indications whether the spaceflight environment affects the B cells that respond to antigen. This, in turn, will help us understand the impact of aging on B cell function.

We anticipate that we will obtain several data sets which have relevance to spaceflight and/or have general significance to understanding host immune responsiveness. These data include: 1) Data that show that spaceflight either maintains or alters the basic distribution of B cells based on Igh transcripts; 2) Information that demonstrates how immunization with an immunogen, commonly used in human medicine, alters the basic distribution of B cells based on Igh transcripts in the absence of the spaceflight variable; 3) Determinations how activation of B cells through Tlr9 receptors alters the basic distribution of B cells based on Igh transcripts.

We will ascertain if activation of B cells through Tlr9 receptors can reverse any impact that may occur during spaceflight.

Task Progress:	Our flight investigation was launched on NG-11 out of Wallops on 4/17/19. At the request of NASA, this flight experiment combined experiments 1 and 2 of our original proposal, including: 1) a primary challenge in space, and 2) a primary challenge pre-flight + a secondary challenge in space. Our third flight experiment is currently under consideration for a future flight date. Members of our team participated in the baseline, environmental control and vivarium control vaccinations and tissue collection at Kennedy Space Center (KSC). All of these events were on a 3-4 day delay after similar procedures were performed by the International Space Station (ISS) astronauts. Collected tissues for ground controls were frozen and shipped to Loma Linda University (LLU) and stored in our freezers at -80°C. A subset of the flight blood and spleen samples were returned from the ISS and shipped to LLU to our lab in August 2019. The remaining samples and frozen carcasses of our flight mice were returned and shipped to LLU in January 2020. To date, we have completed the mRNA isolation from our spleen samples. We are currently conducting a preliminary study to decide if we want to do the sequencing according to our original plan via MiSEQ, or with a newer sequencing technology via Nanopore. Nanopore allows us to obtain much longer sequences and the processing is significantly faster. Because we are focusing specifically on the B cell receptor and TT (tetanus toxoid) specific antibody responses, the long sequences could, potentially, greatly improve the specificity of our analysis. This analysis should be completed in April or May. Once complete, we will begin library prep and sequencing of the flight samples. Additionally, we have performed a tetanus toxoid specific IgG/IgA/IgM ELISA on the blood samples from this study. There were some issues with the samples brought back from the ISS. Specifically, there was significant lysing of the red blood cells (RBCs) and we were unable to separate the cells from the plasma in roughly half of the cells. However, despite these difficulties, we were able to detect Ig in the appropriate samples. Very preliminary data suggests that both the CpG adjuvant and a secondary inoculation with the vaccine increased the antibody responses as expected on the ground. Note the difference in scale for the primary vs. secondary response figures. This indicates that under normal circumstances, B cell memory developed and responded as expected. However, both the primary and secondary responses were drastically inhibited after flight. There are several possible reasons for this. Decreases in the efficacy of 1) the adjuvant, 2) antigen presentation, 3) B cell memory development after the primary injection, or 4) overall cell death in B cell counts are all possibilities. The approval of our last flight experiment (where both the primary and secondary inoculations occur in orbit) is crucial to help identify the actual mechanism. Finally, we have worked with the Biospecimen BSP team at Ames and the tissue collection for all RR12 mice. This was originally scheduled to occur over three weeks in March (3/9-3/27) 2020. However, due to the COVID19 situation, this dissection has been delayed until at least the summer of 2020.
Bibliography Type:	Description: (Last Updated: 08/04/2025)
Abstracts for Journals and Proceedings	Rettig TA, Nishiyama NC, Pecaut MJ, Chapes SK. "Bone Marrow Antibody Repertoire Responses after Vaccination in a Mouse Spaceflight Model." Talk presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019